

EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 7B

The relevant notes for this discussion are [Note 7B](#) and [Note 8](#).

1. System identification by means of least squares

Working through this question will help you understand better how we can use experimental data taken from a (presumably) linear system to learn a discrete-time linear model for it using the least-squares techniques you learned in 16A. You will later do this in lab for your robot car.

As you were told in 16A, least-squares and its variants are not just the basic workhorses of machine learning in practice, they play a conceptually central place in our understanding of machine learning well beyond least-squares.

Throughout this question, you should consider measurements to have been taken from one long trace through time.

- (a) Consider the scalar discrete-time system

$$x[i+1] = ax[i] + bu[i] + w[i] \quad (1)$$

Where the scalar state at time i is $x[i]$, the input applied at time i is $u[i]$ and $w[i]$ represents some external disturbance that also participated at time i .

Assume that you have measurements for the states $x[i]$ from $i = 0$ to m and also measurements for the controls $u[i]$ from $i = 0$ to $m - 1$.

Set up a least-squares problem that you can solve to get an estimate of the unknown system parameters a and b .

- (b) What if there were now two distinct scalar inputs to a scalar system

$$x[i+1] = ax[i] + b_1u_1[i] + b_2u_2[i] + w[i] \quad (2)$$

and that we have measurements as before, but now also for both of the control inputs.

Set up a least-squares problem that you can solve to get an estimate of the unknown system parameters a, b_1, b_2 .

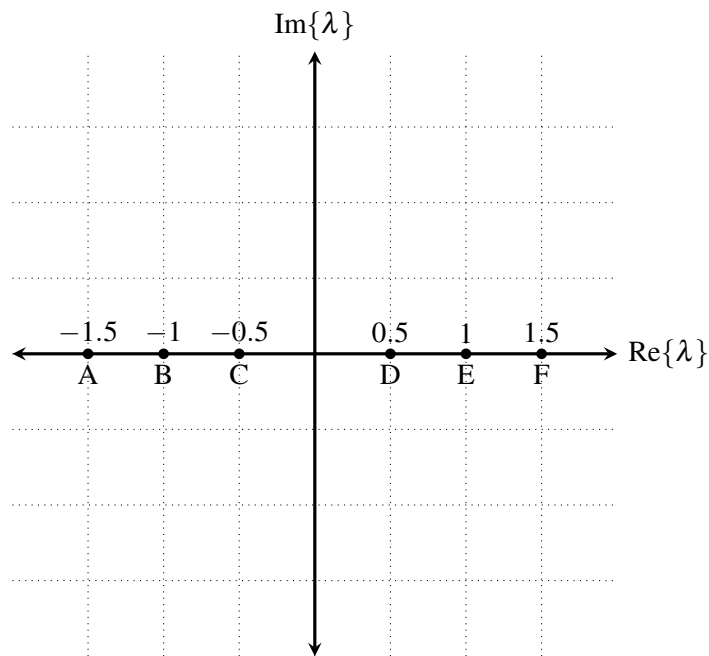
- (c) What could go wrong in the previous case? For what kind of inputs would make least-squares fail to give you the parameters you want?
- (d) Returning to the scalar case with only one input, what could go wrong? When would you be unable to use least-squares to get the parameters you want?
- (e) Now consider the two dimensional state case with a single input.

$$\vec{x}[i+1] = \begin{bmatrix} x_1[i+1] \\ x_2[i+1] \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \vec{x}[i] + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} u[i] + \vec{w}[i] \quad (3)$$

How can we treat this like two parallel problems to set this up using least-squares to get estimates for the unknown parameters $a_{11}, a_{12}, a_{21}, a_{22}, b_1, b_2$? What work/computation can we reuse across the two problems?

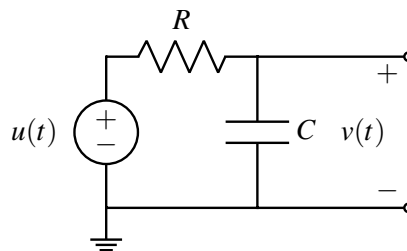
2. Discrete time system responses

We have a system $x[k+1] = \lambda x[k]$. For each λ value plotted on the real-imaginary axis, sketch $x[k]$ with an initial condition of $x[0] = 1$. Determine if each system is stable.



3. Stability Examples and Counterexamples

- (a) Consider the circuit below with $R = 1\Omega$, $C = 0.5\text{F}$, and $u(t) = \cos(t)$. Furthermore assume that $v(0) = 0$ (that the capacitor is initially discharged).



This circuit can be modeled by the differential equation

$$\frac{d}{dt}v(t) = -2v(t) + 2u(t) \quad (4)$$

Show that the differential equation is always stable. Consider what this means in the physical circuit.

- (b) Consider the discrete system

$$x[k+1] = 2x[k] + u[k] \quad (5)$$

with $x[0] = 0$.

Is the system stable or unstable? If unstable, find a bounded input sequence $u[k]$ that causes the system to “blow up”. If unstable, is there still a (non-trivial) bounded input sequence that does not cause the system to “blow up”?

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